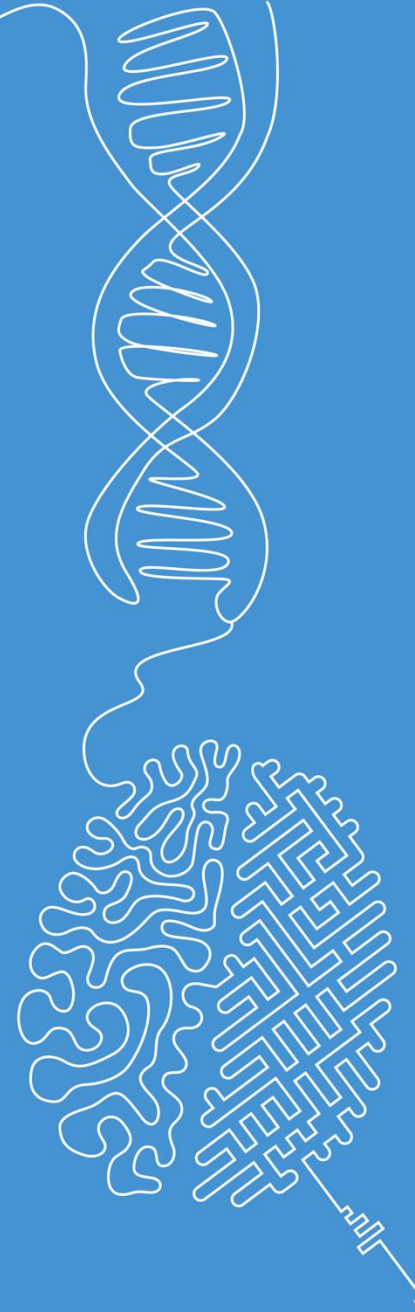


Fourier Transformation

Image and Signal Processing

Norman Juchler



Fourier transformation – Outline

- **Idea:** Treat the aperiodic function $x(t)$ like a periodic function with period $T \rightarrow \infty$ and then compute Fourier series. In the limit, this results in the Fourier transform.

- **Fourier transform**

$$\hat{x}(k) = \int_{-\infty}^{\infty} x(t) e^{-ikt} dt \quad k \in \mathbb{R}$$

- **Inverse Fourier transform**

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{x}(k) e^{ikt} dk$$

For comparison: the Fourier series formulas

$$c_k = \frac{1}{T} \int_0^T x(t) \cdot e^{-i2\pi \frac{k}{T} t} dt$$

$\forall k \in \mathbb{Z}$

$$x(t) = \sum_{k=-N}^N c_k e^{\frac{2\pi i k}{T} t}$$

...do you see similarities?

- The Fourier transform generalizes the Fourier series to aperiodic signals. It allows us to analyze signals that do not repeat periodically.
- Disclaimer: The above integrals exists for many, but not all, signals. Here, we skip the technical details.

Properties of Fourier Transforms

Properties

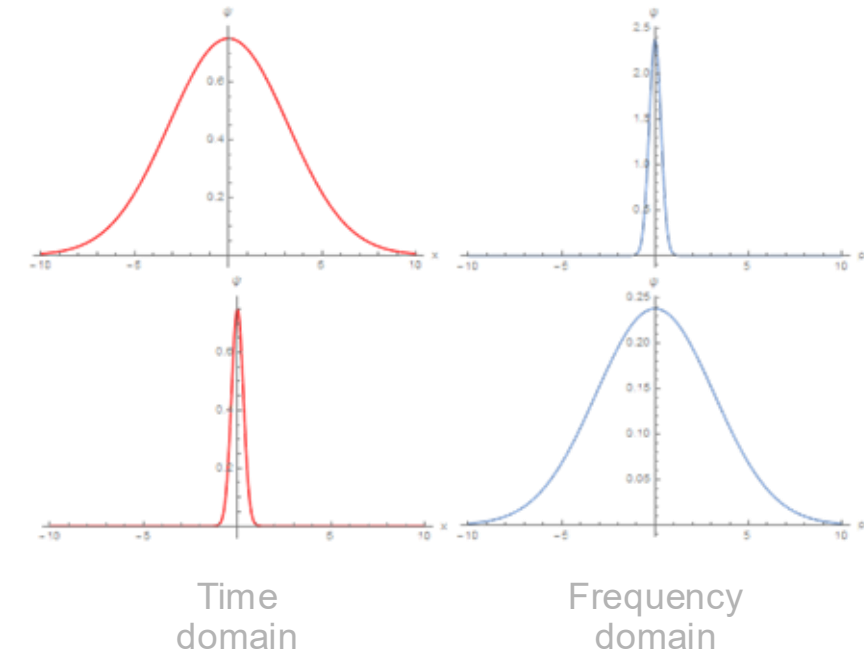
The Fourier transform exhibits many unexpected and beautiful properties. Here is a selection:

- **Uncertainty principle:** A function that is narrow in the time domain becomes broad in the frequency domain, and vice versa.

$$x(at) \Leftrightarrow \frac{1}{a} X\left(\frac{\omega}{a}\right)$$

- **Linearity:** A linear combination of functions in the time domain corresponds to the same linear combination of their Fourier transforms in the frequency domain.

$$a \cdot x(t) + b \cdot y(t) \Leftrightarrow a \cdot X(\omega) + b \cdot Y(\omega)$$



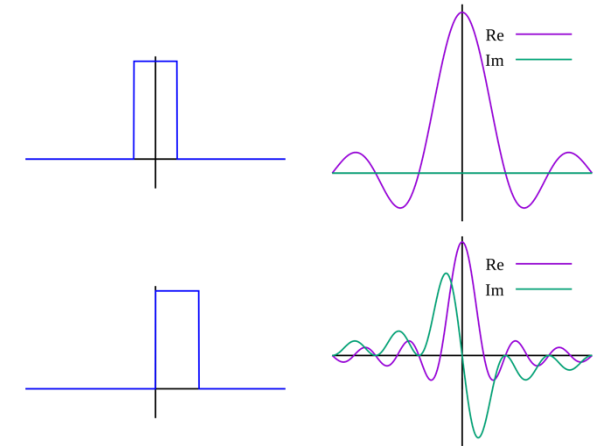
Properties

- **Time shift:** Shifting a signal $x(t)$ by t_0 introduces a linear phase term in its Fourier transform. The magnitude remains unchanged:

$$x(t - t_0) \Leftrightarrow X(\omega) \cdot e^{-i\omega t_0}$$

- **Differentiation** (in time domain): The differentiation of a function in time domain is equivalent to the multiplication of its Fourier transform by a factor $i\omega$ in frequency domain

$$\frac{d}{dt}x(t) \Leftrightarrow (i\omega) \cdot X(\omega)$$



We've seen this effect already for the Fourier series of a rectangular wave

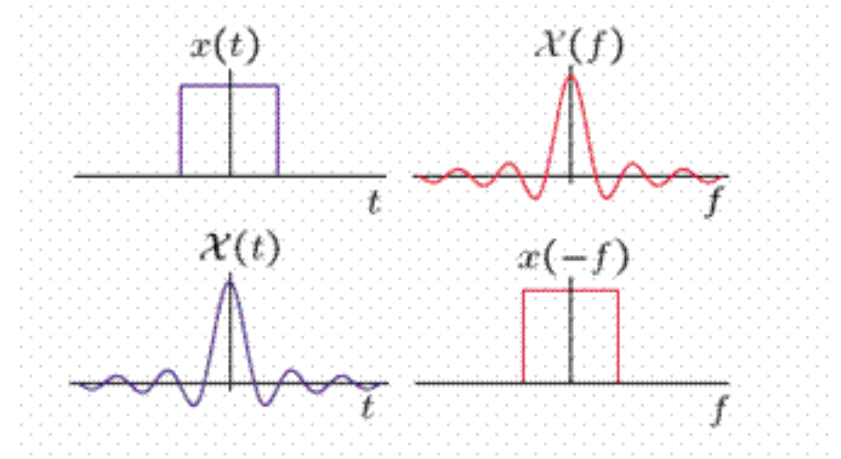
Illustration of why a time shift affects only the phase and not the magnitude:

$$\begin{aligned} X(\omega) \cdot e^{-i\omega t_0} &= |X(\omega)| \cdot e^{-i\omega\varphi(\omega)} \cdot e^{-i\omega t_0} \\ &= |X(\omega)| e^{-i\omega(\varphi(\omega) + t_0)} \end{aligned}$$

Properties

- **Duality:** If $x(t)$ has a Fourier transform $X(\omega)$, and we construct a new time-domain function using the shape of $X(\omega)$, then its Fourier transform will have the shape of $x(t)$ but as a function of frequency. (Time $\leftrightarrow$ frequency roles are exchanged.)

$$x(t) \leftrightarrow X(t) \quad \Leftrightarrow \quad X(t) \leftrightarrow 2\pi x(-\omega)$$



- **Energy preserving:** The total energy in a signal across all time is equal to the total energy in the transform across all frequencies (Parseval's theorem)

$$\int_{\mathbb{R}} |x(t)|^2 dt = \int_{\mathbb{R}} |X(\omega)|^2 d\omega$$

Properties

- **Symmetry:** From the sine–cosine form, we can derive the following properties based on the symmetry of the input function.

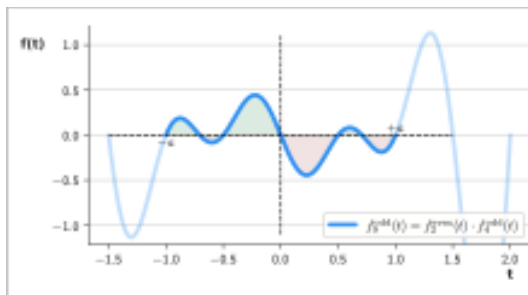
$$\hat{x}(\omega) = X(\omega) = \int_{-\infty}^{\infty} x(t) \cos(\omega t) dt - i \int_{-\infty}^{\infty} x(t) \sin(\omega t) dt$$

- If $x(t)$ is real-valued and even, the FT is a real-valued function!
- If $x(t)$ is real-valued and odd, the FT is imaginary-valued!

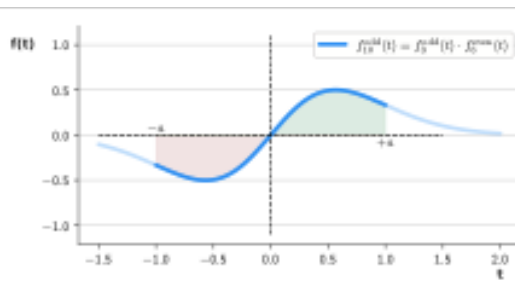
$$X(\omega) = R(\omega t)$$

$$X(\omega) = I(\omega t)$$

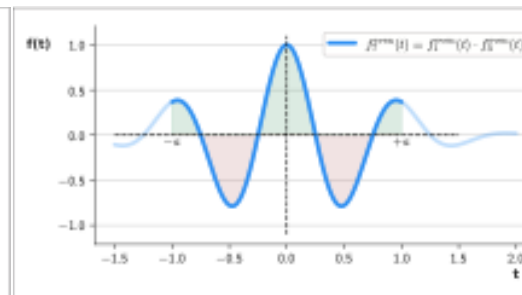
Even × Odd = Odd



Odd × Even = Odd



Even × Even = Even



Odd × Odd = Even

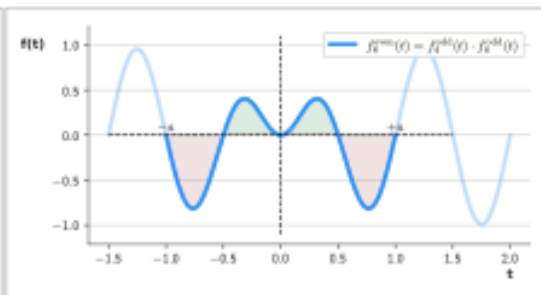


Illustration of how odd and even functions interact with cosine (even) and sine (odd).
Note: over symmetric limits, an odd integrand integrates to zero.

Fourier transform pairs

- As with Fourier series, tables of common Fourier transforms are available for frequently used functions.

Table of Common Functions and their Fourier Transforms

Function name	Function in the time domain	Fourier Transform (in the frequency domain)
	$w(t)$	$\hat{W}(f)$
Dirac delta	$\delta(t)$	1
Constant	1	$\delta(f)$
Cosine	$\cos(2\pi f_0 t)$	$\frac{\delta(f - f_0) + \delta(f + f_0)}{2}$
Sine	$\sin(2\pi f_0 t)$	$\frac{\delta(f - f_0) - \delta(f + f_0)}{2j}$
Unit step function	$u(t) = \begin{cases} 0, & \text{if } t < 0 \\ 1, & \text{if } t \geq 0 \end{cases}$	$\frac{1}{j\omega}$ (for $\omega = 2\pi f$)
Decaying exponential (for $t > 0$)	$e^{-\alpha t}u(t)$,	$\frac{1}{\alpha + j2\pi f}$, $\alpha > 0$
Box or rectangle function	$\text{rect}(at) = \begin{cases} 0, & \text{if } at > \frac{1}{2} \\ 1, & \text{if } at \leq \frac{1}{2} \end{cases}$	$\frac{1}{ a } \text{sinc}\left(\frac{f}{a}\right) = \frac{\sin(\pi f/a)}{\pi f/a}$
Sinc function	$\text{sinc}(at) = \frac{\sin(\pi at)}{\pi at}$	$\frac{1}{ a } \text{rect}\left(\frac{f}{a}\right)$
Comb function	$\sum_{n=-\infty}^{\infty} \delta(t - nT)$	$\frac{1}{T} \sum_{k=-\infty}^{\infty} \delta\left(f - \frac{k}{T}\right)$
Gaussian	$e^{-\alpha t^2}$	$\sqrt{\frac{\pi}{\alpha}} e^{-\frac{(\pi f)^2}{\alpha}}$

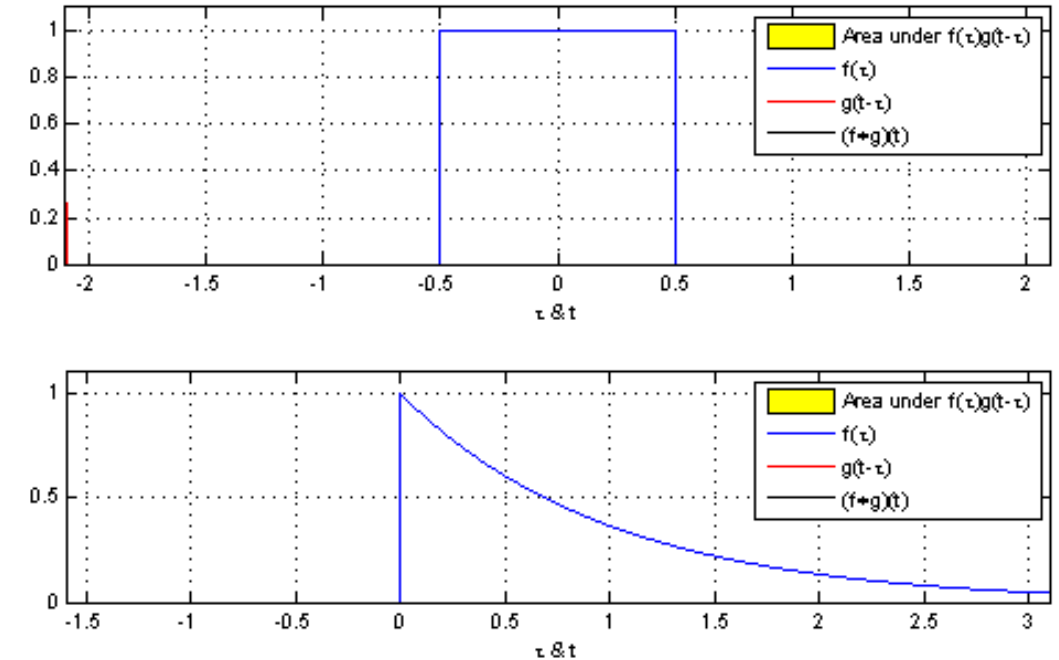
Convolution

Convolution

- We still need one more key property of the Fourier transform.
- To introduce it, we first define a new concept: the convolution of two signals.

$$y(t) = (h * x)(t) := \int_{-\infty}^{\infty} h(\tau)x(t - \tau) d\tau$$

- The convolution combines two functions to produce a third function.
- It computes the integral of the product $h(\tau)$ and $x(t - \tau)$, as τ varies.

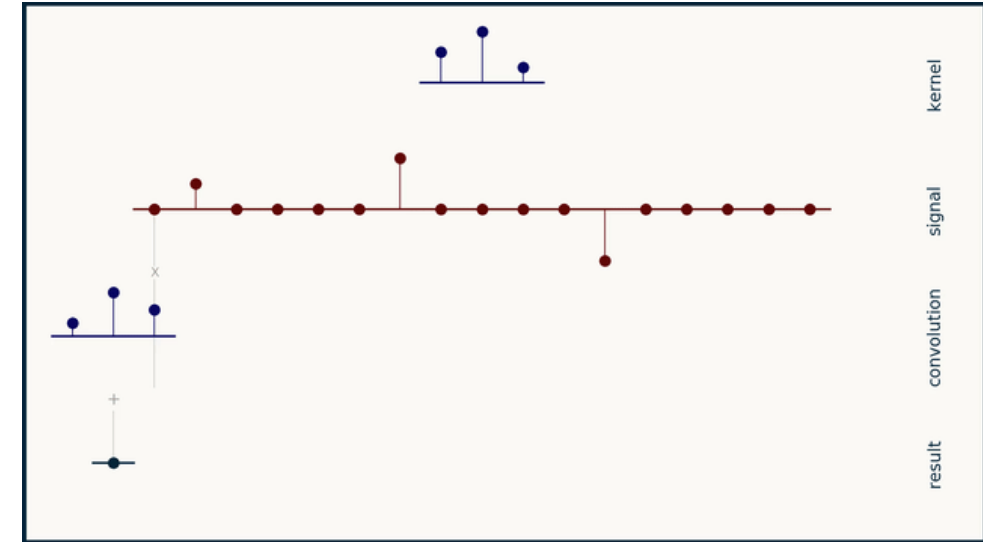


Convolution

- There exists also a discrete-time definition for convolution:

$$y[n] = (h * g)[n] = \sum_{m=-\infty}^{\infty} h[m]x[n-m]$$

- In the animation on the right, $x[n]$ is the input signal, $h[n]$ the kernel, and $y[n]$ the output signal.



Convolution and Fourier transform

- **Result:** Result: The Fourier transform reduces convolution to simple multiplication in the frequency domain.

$$y(t) = h(t) * x(t) \quad \Leftrightarrow \quad H(\omega) \cdot X(\omega)$$

- Because convolution in the time domain is comparatively difficult to compute, converting the signals with the Fourier transform, multiplying them in the frequency domain, and then applying the inverse Fourier transform is often much easier.
- This is especially powerful for discrete-time signals when combined with the FFT.
- Convolution also forms the foundation of many digital signal filters — more on this in upcoming lectures.